Log-Normal Tail Probabilities

Example. Consider a non-dividend-paying stock. What is the probability that the stock outperforms a risk-free investment under the risk-neutral probability measure?

$\longrightarrow$: The initially invested amount is $S(0)$.

- If it's a risk-free investment, the balance @ time T is $\underline{S(0)e^{rT}}$.
- If it's a stock investment, the wealth @ time T is $\underline{S(T)}$.

$$\mathbb{P}^*\left[S(T) > S(0)e^{rT}\right] = ?$$

This question is equivalent to the one of whether the profit for the stock investment is positive under $\mathbb{P}^*$

In the Black-Scholes model:

$$S(T) = S(0)e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} \quad \text{w/ } Z \sim N(0,1)$$

$$\mathbb{P}^*\left[\cancel{S(0)}\, e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} > \cancel{S(0)}\, e^{rT}\right] =$$

$\ln(\quad)$ $\ln(\quad)$ $\ln(\cdot)$ is increasing

$$= \mathbb{P}^*\left[(\cancel{r} - \frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z > \cancel{rT}\right]$$

$$= \mathbb{P}^*\left[-\frac{\sigma^2}{2}\cdot T + \sigma\sqrt{T}\cdot Z > 0\right]$$

$$= \mathbb{P}^*\left[\sigma\sqrt{T}\cdot Z > \frac{\sigma^2}{2}\cdot T\right]$$

$$= \underline{\mathbb{P}^*\left[Z > \frac{\sigma\sqrt{T}}{2}\right]}$$

$$= \mathbb{P}^*\left[Z < -\frac{\sigma\sqrt{T}}{2}\right] = N\left(-\frac{\sigma\sqrt{T}}{2}\right) \xrightarrow[T\to\infty]{} 0 \quad \square$$

Example. Consider a European call w/ strike K and exercise date T.
Under the risk-neutral probability measure, what is the probability that the call is in-the-money @ time T?

$\longrightarrow$: $\mathbb{P}^*[S(T) > K] =$

$$= \mathbb{P}^*\left[S(0)\, e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} > K\right] =$$

$$= \mathbb{P}^*\left[e^{(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z} > \frac{K}{S(0)}\right]$$

$\ln(\cdot)$ increasing

$$= \mathbb{P}^*\left[(r-\frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z > \ln\left(\frac{K}{S(0)}\right)\right]$$

$$= \mathbb{P}^*\left[\sigma\sqrt{T}\cdot Z > \ln\left(\frac{K}{S(0)}\right) - (r-\frac{\sigma^2}{2})\cdot T\right]$$

$$= \mathbb{P}^*\left[Z > \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{K}{S(0)}\right) - (r-\frac{\sigma^2}{2})\cdot T\right)\right]$$

symmetry of $N(0,1)$

$$= \mathbb{P}^*\left[Z \leq \boxed{\frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{S(0)}{K}\right) + (r-\frac{\sigma^2}{2})\cdot T\right)}\right]$$

$=: d_2$

$$\boxed{\mathbb{P}^*[S(T) > K] = N(d_2)}$$

$\square$

Consequently.
The probability under $\mathbb{P}^*$ that the otherwise identical put is in-the-money @ time T is

$$\mathbb{P}^*[S(T) < K] = 1 - N(d_2) = N(-d_2)$$

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Problem Set 14

Black-Scholes pricing.

Problem 14.1. Let the stock prices be modeled using the lognormal distribution. Under the risk-neutral probability measure, the mean stock price at time−1 equals 120 and the median stock price 115. What is the risk-neutral probability that the time−1 stock price exceeds 100?

$\rightarrow$:

$$\mathbb{P}^*[S(1) > 100] = \cancel{X}$$

$$\mathbb{P}^*\left[S(0)e^{(r-\frac{\sigma^2}{2})\cdot 1 + \sigma\sqrt{1}\cdot Z} > 100\right] =$$

$$= \mathbb{P}^*\left[\underbrace{\boxed{S(0)e^{(r-\frac{\sigma^2}{2})}}}_{\text{median of } S(1)\ =\ 115} \cdot e^{\sigma\cdot Z} > 100\right] =$$

$$= \mathbb{P}^*\left[115\, e^{\sigma\cdot Z} > 100\right] = \mathbb{P}^*\left[e^{\sigma\cdot Z} > \frac{100}{115}\right]$$

$$= \mathbb{P}^*\left[Z > \frac{1}{\boxed{\sigma}}\ln\left(\frac{100}{115}\right)\right] =$$

$\boxed{\sigma} = ?$

$$\frac{\text{mean}}{\text{median}} = \frac{120}{115} = e^{\frac{\sigma^2}{2}\cdot 1} \Rightarrow \sigma = \sqrt{2\cdot\ln\left(\frac{120}{115}\right)} = \underline{0.2918}$$

$$\mathbb{P}^*[Z > -0.48] = N(0.48) = \underline{0.6844}$$

$\square$